

and a robust punch in the audio, which is enhanced by excellent spatial rendering that offers a wide stereo sound and accurate distance portrayal. The 14-inch variant continues to feature a six-speaker array with force-cancelling woofers and support for Dolby Atmos, maintaining its reputation for “studio-quality” performance.

The speakers are full-bodied and bassy, providing an immersive audio experience for a range of uses, from music listening to video calls and movie watching. I also tested the 15-inch MacBook Air earlier this year and it also includes a six-speaker system, an upgrade from the four speakers in the smaller 13-inch model. This configuration comprises two dedicated tweeters and two sets of force-cancelling woofers, effectively doubling the bass response compared to the 13-inch model.

While the audio quality of the 15-inch MacBook Air is an improvement over its smaller counterpart, it falls slightly short of the richness and depth offered by the 14-inch MacBook Pro. The Air model ejects audio from behind the keyboard and against the display, and though it supports Dolby Atmos Spatial Audio, the effect isn't as pronounced as in the Pro model. Overall, while both models provide high-quality sound experiences, the 14-inch MacBook Pro stands out for its richer and more immersive audio performance, setting a higher benchmark.

The MacBook Pro 14-inch features a Liquid Retina XDR display, which is a 14.2-inch diagonal screen with a native resolution of 3024-by-1964 at 254 pixels per inch. It can deliver 1000 nits sustained full-screen and a peak of 1600 nits for HDR content, while SDR brightness is at 600 nits, up from 500 nits from the previous gen.

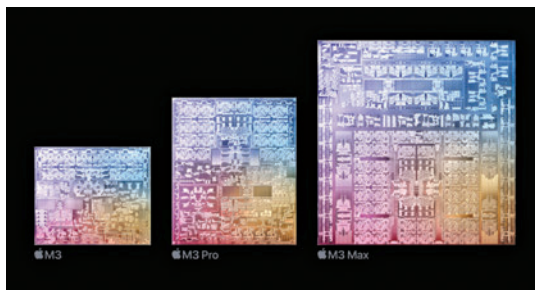
It covers 100% of the DCI-P3 colour gamut, which is widely used in digital cinema. It also offers excellent colour accuracy, making it suitable for professional colour grading and graphic design work. The display supports over 1 billion colours, wide colour (P3), and True Tone technology. Additionally, it has ProMotion technology for adaptive refresh rates up to 120Hz, alongside fixed refresh rates

including 47.95Hz, 48.00Hz, 50.00Hz, 59.94Hz, and 60.00Hz. The display is fantastically bright, particularly when viewing HDR content, and has smooth scrolling thanks to a 120Hz ProMotion refresh rate.

## Gaming And Benchmark Performance

It doesn't matter if you're playing on battery or when this is plugged in because in both cases, the MacBook's performance is absolutely the same. And it is something which you hardly ever see in any Windows laptop and I really appreciate it.

At this point, more than 34% of the M3 Max chip area is designated for GPU. And it is a fairly powerful one. It features hardware ray tracing and mesh shaders,



aligning with DX12 feature level 2. Here are some numbers to help you understand how capable this chip is. First up is the Geekbench 6 GPU test in OpenCL. And it scored as well as an Nvidia RTX 4070 GPU powering the Samsung Galaxy Book3 Ultra, albeit a bit less.

Moving on, in 3D Mark's Wildlife the M3 Max scored the maximum score possible. Meanwhile, in Wildlife Extreme it scored 20400 compared to 12423 scored by the Samsung Galaxy Book3 Ultra. The score difference is pretty huge here because this benchmark is more focused on 3D graphics rendering, a task that can benefit significantly from specific GPU architectural features like efficient handling of textures and shaders, something we've seen the M3 Max can handle really well because of new hardware features.

Now, let's talk about the CPU in the M3 Max SoC. I understand that online discourse has claimed that the 16-core CPU throttles quite a lot in its operations.

So, does that statement hold? And if it does, what kind of performance drop are we actually expecting? Here's an answer.

During the Cinebench R23 single loop, the 16-C CPU scores the highest and unsurprisingly, it scores 21258 during the 10-minute test loop and 21264 during a 30-minute test loop. So the M3 Max chip does throttle and the point difference is more than 2700 points, which is really not good. A deeper look at this test reveals that the maximum TGP drawn by the M3 Max begins to fall halfway through the test, most notably when the CPU core temperature crosses 102 degrees Celsius. This clearly shows that the 14-inch MacBook Pro chassis is struggling to let the M3 Max stretch its legs at full speed for a long duration.

During normal operations, like video streaming, browsing the web, editing documents etc, the fans don't even get activated. However, during gaming, the 14-inch chassis gets too hot, especially in the middle of the keyboard, and then the entire system begins to throttle. Moreover, the

resulting fan noise is too high as well, reaching close to 7200 RPM during benchmarks. And you can see the same result while gaming as well. I tested Lies of P and the fan speed was at its maximum as well along with the CPU temperature. After a thorough gaming test, here's the result.

So all these games can run at 1080p at high settings and still deliver either more than or around 100 FPS. And while running all these titles, the max fan speed was still at its peak and the laptop surface temperature was also around 46-47 degrees Celsius. Making this laptop too loud and hot during gameplay, the latter will bother you more provided you're playing using the keyboard and not a gamepad.

In short, even though impressive, this is still not the maximum performance the M3 Max chip can deliver. It still has room to stretch its legs but is limited by the restricted cooling solution which is a result of a smaller 14-inch chassis. **d**